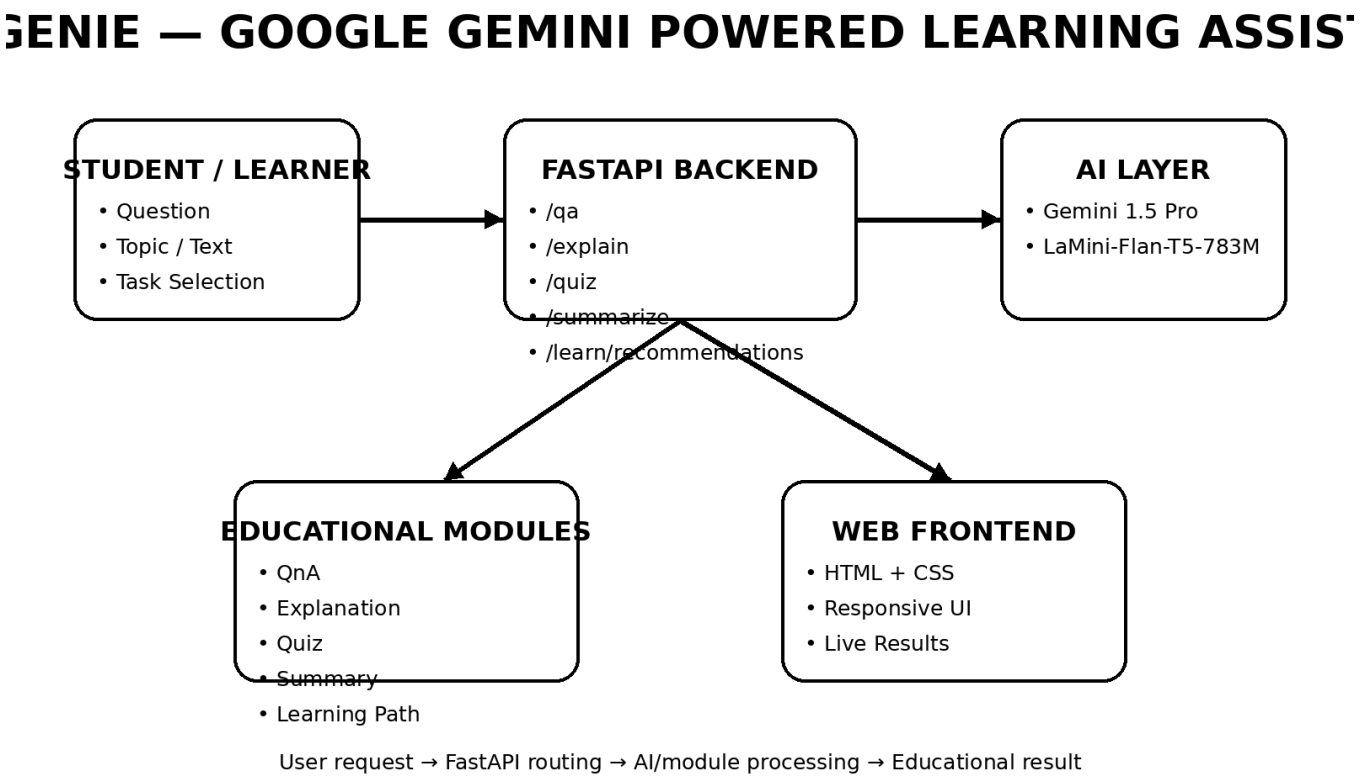
EDUGENIE

Google Gemini Powered Learning Assistant



Project Report

Prepared from the supplied EduGenie project document and the supplied brainstorming/ideation reference.

1. Project Overview & Ideation

EduGenie is described in the supplied project document as a lightweight AI-powered educational assistant designed to simplify learning through generative AI. It supports question answering, simplified explanations, quiz generation, summarization, and personalized learning recommendations.

The supplied brainstorming reference uses a problem-identification → core-idea → user-journey → design-principles structure. Following that structure, EduGenie can be presented as a learner-focused web application that turns a student's question or study material into an interactive learning response.

Problem Identification

Students may need different forms of help from the same study material: a direct answer, a simpler explanation, a quiz for self-checking, a concise summary, or a structured path for learning a topic. EduGenie combines these functions in one lightweight interface.

Core Idea

Build a web-based learning assistant using FastAPI for the backend and HTML/CSS for the frontend, with Google Gemini for major generative-AI tasks and LaMini-Flan-T5-783M for concept explanation as described in the supplied document.

Proposed User Journey

1. Open EduGenie → 2. Select a task → 3. Enter a question/topic/text → 4. Submit → 5. Backend processes the request → 6. AI/module generates the result → 7. Result is displayed in the web interface.

Design Principles

The source emphasizes a simple web interface, modular backend files, lightweight infrastructure, accessible learning support, and separate modules for explanation, Q&A;, quizzes, summaries, and learning recommendations.

2. Objectives and Key Features

Objectives

• Provide quick question-and-answer support for academic and general-knowledge queries.
• Explain concepts in simpler language for learners.
• Generate multiple-choice quizzes from supplied content.
• Summarize long educational passages for revision.
• Generate structured learning recommendations from beginner to advanced levels.

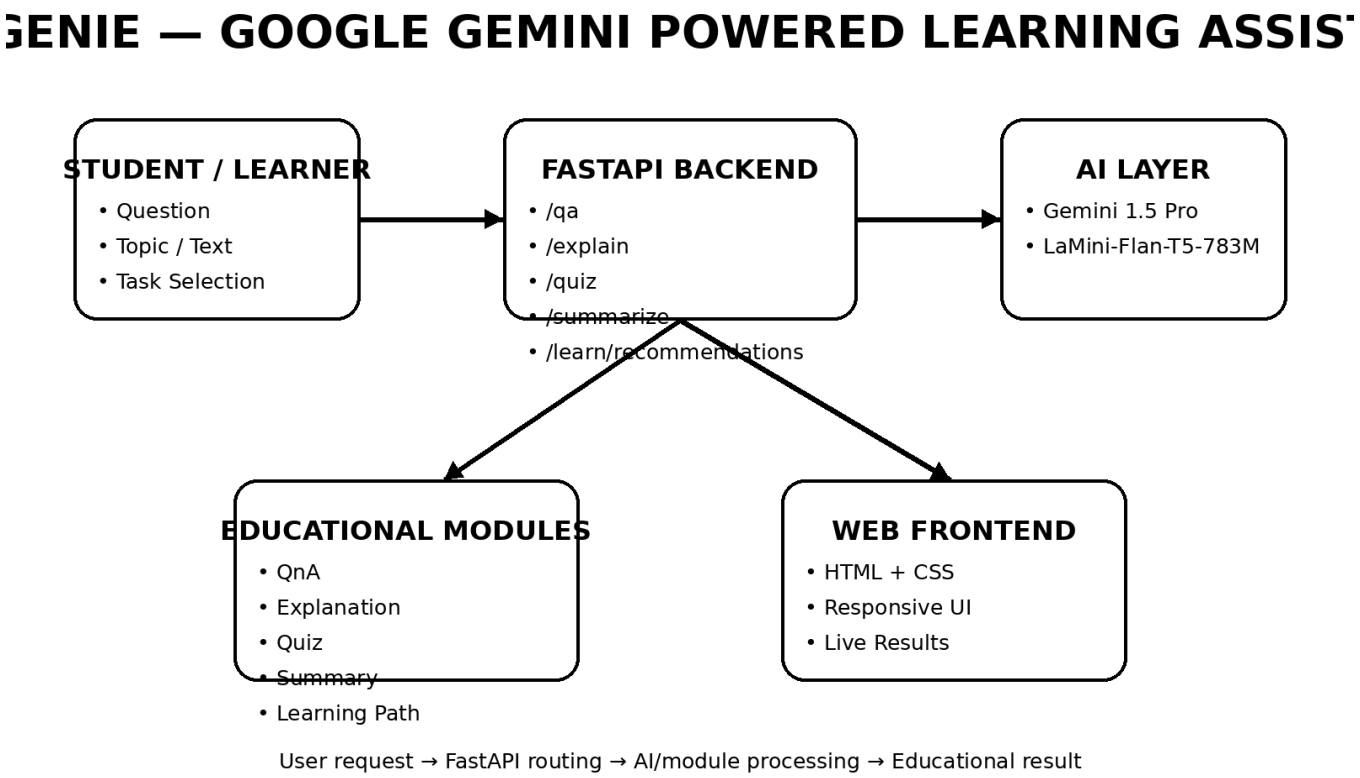
Key Features

Feature	Purpose
Q&A	Answer learner questions using the Gemini-based QnA module.
Explanation	Break down concepts into concise, readable explanations.
Quiz	Generate three MCQs with four options each from a passage.
Summary	Convert longer educational content into a concise revision-oriented summary.
Learning Path	Suggest a structured beginner-to-advanced learning path with resources.

Scenarios

The supplied document gives examples such as asking for the largest ocean, generating a quiz to check understanding of the Pythagoras Theorem, and requesting a structured learning path for SQL.

3. System Architecture



The architecture follows the source document: a simple HTML/CSS frontend communicates with a FastAPI backend. REST endpoints route requests to dedicated educational modules, which use the documented AI models to generate responses.

Project Folder Architecture

EduGenie/
main.py — FastAPI application
explanation_module.py — concept explanation logic
qna.py — question answering
quiz_module.py — quiz generation
summary_module.py — summarization
learning_path.py — learning recommendations
templates/index.html — HTML frontend
static/style.css — styling
requirements.txt — Python dependencies

4. AI Modules and API Endpoints

AI Model Selection

According to the supplied project document, Gemini 1.5 Pro via API is used for Q&A;, summarization, quiz generation, and learning paths. LaMini-Flan-T5-783M is used locally for concept explanation.

Module Responsibilities

Module	Documented responsibility
Explanation Module	Uses LaMini-Flan-T5-783M to provide simplified, readable explanations.
QnA Module	Uses Gemini 1.5 Pro for general knowledge and academic question answering.
Quiz Module	Uses Gemini to generate three MCQs, four options each, with a correct answer in structured JSON.
Summary Module	Uses Gemini to condense long paragraphs while retaining core information.
Learning Path Module	Uses Gemini to generate structured beginner-to-advanced learning recommendations and resources.

REST API Endpoints

/qa
/explain
/quiz
/summarize
/learn/recommendations

The source states that each endpoint is connected to its corresponding module logic, enabling browser use and functional testing.

5. Frontend, Workflow and Deployment

Frontend

The documented frontend uses an HTML form with a task dropdown containing Explain, QnA, Quiz, Summary, and Recommend Path; a text area for user input; and a submit button. CSS provides responsive styling for inputs, buttons, and the result area.

Live Integration Workflow

1. User selects a learning task.
2. User enters a question, topic, or passage.
3. The browser sends a POST request to the FastAPI backend.
4. The selected backend endpoint invokes the corresponding module.
5. The AI/model generates the requested learning output.
6. The result is displayed below the input area.

Local Deployment

The supplied document specifies running the application locally with:

uvicorn main:app --reload

and accessing it at **http://127.0.0.1:8000**.

Prerequisites documented in the source include Python 3.10+, FastAPI, HTML/CSS, a Google Gemini API key, Uvicorn, and Jinja2.

6. Functional Testing and Conclusion

Functional Testing

The project document lists functional checks for asking a question, getting an explanation, generating a quiz, summarizing content, and obtaining a personalized learning plan. The quiz workflow is described as generating three questions with four options each and indicating/correcting an incorrect selection.

Expected Project Outcome

EduGenie is presented in the source as an accessible AI-powered educational assistant that combines cloud-based generative AI with a lightweight web interface. Its modular design supports different learning tasks while keeping the frontend simple.

Conclusion

Based on the supplied documentation, EduGenie provides a unified learning interface for Q&A;, concept explanation, quiz generation, summarization, and learning recommendations. The documented architecture separates the web interface, FastAPI routes, and individual AI modules, making the application suitable for demonstration, testing, and future extension.

Reference basis: supplied EduGenie project document and supplied brainstorming/ideation PDF.